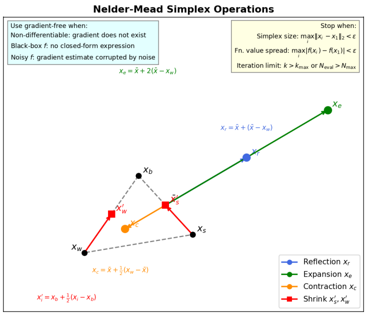

col 3 placeholder

col 4 placeholder

(h) [3 marks] The particle swarm update is:

$$x_{k+1}^{(i)} = x_k^{(i)} + v_{k+1}^{(i)} \Delta t, \qquad v_{k+1}^{(i)} = \alpha v_k^{(i)} + \beta \frac{x_{\text{best}}^{(i)} - x_k^{(i)}}{\Delta t} + \gamma \frac{x_{\text{best}} - x_k^{(i)}}{\Delta t}.$$

Define all symbols.

Solution:

- $x_k^{(i)}$: position of particle i at iteration k
- $v_k^{(i)}$: velocity of particle i at iteration k
- $x_{\text{best}}^{(i)}$: best position ever visited by particle i (personal best)
- x_{best}: best position ever found by *any* particle in the swarm (global best)
- α: inertia weight (momentum of previous velocity)
- β: cognitive coefficient (attraction to personal best)
- γ: social coefficient (attraction to global best)
- Δt: time step; k = iteration index; i = particle index